

INDR 252 Midterm 2 Study Questions

1)

A certain type of stainless steel powder is supposed to have a mean particle diameter of $\mu = 15 \mu\text{m}$. A random sample of 87 particles had a mean diameter of $15.2 \mu\text{m}$, with a standard deviation of $1.8 \mu\text{m}$. A test is made of $H_0 : \mu = 15$ versus $H_1 : \mu \neq 15$.

- Find the P -value.
- Do you believe it is plausible that the mean diameter is $15 \mu\text{m}$, or are you convinced that it differs from $15 \mu\text{m}$? Explain your reasoning.

Solution:

$$n_x = 87 \quad \bar{X} = 15.2 \quad s_x = 1.8$$

$$H_0: \mu = 15 \quad \text{versus} \quad H_1: \mu \neq 15$$

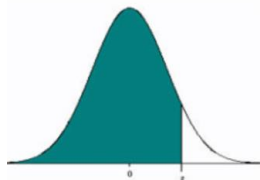
a)

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{s_x / \sqrt{n_x}} = \frac{15.2 - 15}{1.8 / \sqrt{87}} = 1.036375 \rightarrow \text{(two tailed)}$$

$$P\text{-value} = 2 * P(Z > 1.04) = 2 * (1 - P(Z < 1.04)) = 2 * (1 - 0.8508) = 0.2984$$

Table of Standard Normal Probabilities for Positive Z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817



b) If the mean diameter were 15, the probability of observing a sample with a mean as extreme as or more extreme than that observed (15.2) is 0.2984. Since 0.2984 is not a very small probability, we do not reject the null hypothesis. Instead, we conclude that the null hypothesis is plausible, and we cannot reject it.

2)

A new concrete mix is being designed to provide adequate compressive strength for concrete blocks. The specification for a particular application calls for the blocks to have a mean compressive strength μ greater than 1350 kPa. A sample of 100 blocks is produced and tested. Their mean compressive strength is 1356 kPa and their standard deviation is 70 kPa. A test is made of $H_0 : \mu \leq 1350$ versus $H_1 : \mu > 1350$.

- a. Find the P -value.
- b. Do you believe it is plausible that the blocks do not meet the specification, or are you convinced that they do? Explain your reasoning.

Solution:

$$n_x = 100 \quad \bar{X} = 1356 \quad s_x = 70$$

$$H_0: \mu \leq 1350 \quad \text{versus} \quad H_1: \mu > 1350$$

a)

$$Z = \frac{\bar{X} - \mu_0}{s_x / \sqrt{n_x}} = \frac{1356 - 1350}{70 / \sqrt{100}} = 0.857 \approx 0.86 \rightarrow \quad (\text{one tailed}) \quad P - \text{value} = 0.1949$$

b)

We cannot reject the null hypothesis since p -value is high, and $\mu \leq 1350$ is plausible.

3)

If $P = 0.50$, which is the best conclusion?

- i. H_0 is definitely false.
- ii. H_0 is definitely true.
- iii. There is a 50% probability that H_0 is true.
- iv. H_0 is plausible, and H_1 is false.
- v. Both H_0 and H_1 are plausible.

Solution:

- i) False, when $P = 0.5 \rightarrow \bar{X} = \mu_0$ and the null hypothesis is definitely plausible (we cannot reject the null hypothesis).
- ii) False, not being able to reject the null hypothesis means it might be true or false.
- iii) False, P-value is as probability of observing a value of a statistics such as $\bar{X}$ when H_0 is true. The P-value can change, but the truth or falsehood of the null hypothesis cannot be changed by repeating the experiment.
- iv) False, we cannot reject the null hypothesis; therefore, we cannot accept or reject the alternative hypothesis!
- v) True.

4)

A machine that fills cereal boxes is supposed to be calibrated so that the mean fill weight is 12 oz. Let μ denote the true mean fill weight. Assume that in a test of the hypotheses $H_0 : \mu = 12$ versus $H_1 : \mu \neq 12$, the P -value is 0.30.

- a. Should H_0 be rejected on the basis of this test? Explain.
- b. Can you conclude that the machine is calibrated to provide a mean fill weight of 12 oz? Explain.

Solution:

a)

For $\alpha = 5\%$, observing a p-value of 0.30 is not small enough to reject the null hypothesis, since $\alpha = 5\% < p\text{-value} = 30\%$. Considering that p-value is the probability of observing a test statistic as extreme as or more extreme than the observed value, assuming that the null hypothesis is true; we do not have enough evidence against the null hypothesis.

b)

for all reasonable α we can conclude that the machine is calibrated to provide a mean fill weight of 12 oz.

5)

A grinding machine will be qualified for a particular task if it can be shown to produce less than 8% defective parts. In a random sample of 300 parts, 12 were defective. On the basis of these data, can the machine be qualified?

Solution:

Let p denote the probability that a randomly chosen part will be defective.

$$n_x = 300 \quad X = 12 \quad \hat{p}_x = \frac{12}{300} = 0.04$$

$$H_0: p \geq 0.08 \quad \text{versus} \quad H_1: p < 0.08$$

Under null hypothesis $\hat{p}$ is normally distributed with mean 0.08 and standard deviation

$$\sqrt{\frac{0.08(1-0.08)}{300}} = 0.01566312$$

$$Z = \frac{0.04 - 0.08}{0.01566312} = -2.55377 \approx -2.55 \rightarrow P\text{-value} = 0.0054$$

We can reject null hypothesis for all reasonable α values, therefore we conclude that $p < 0.08$.

6)

A certain manufactured product is supposed to contain 23% potassium by weight. A sample of 10 specimens of this product had an average percentage of 23.2 with a standard deviation of 0.2. If the mean percentage is found to differ from 23, the manufacturing process will be recalibrated.

- State the appropriate null and alternate hypotheses.
- Compute the P -value.
- Should the process be recalibrated? Explain.

Solution:

$$n_x = 10 \quad \bar{X} = 23.2\% \quad s_x = 0.2$$

a)

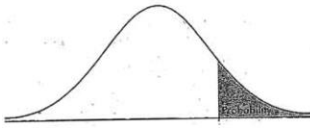
$$H_0: \mu = 23\% \quad \text{versus} \quad H_1: \mu \neq 23\%$$

b)

$$t = \frac{\bar{X} - \mu}{s_x / \sqrt{n_x}} = \frac{23.2 - 23}{0.2 / \sqrt{10}} = 3.1623 \quad n - 1 = 9 \quad (\text{two-tailed}) \rightarrow$$

TABLE B: t-DISTRIBUTION CRITICAL VALUES

df	Tail probability p										
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646



From the table, we find that the area under the curve in each tail is between 0.01 and 0.005. Since this is a two-tailed test, the P-value is the sum of the areas in both tails. We conclude that $0.01 < P\text{-value} < 0.02$.

c)

If $\alpha \geq 2\%$, then we can reject the null hypothesis, and the process should be recalibrated. If $\alpha \leq 1\%$, then we cannot reject the null hypothesis, and recalibration is not necessary.

Z

In a certain supermarket, a sample of 60 customers who used a self-service checkout lane averaged 5.2 minutes of checkout time, with a standard deviation of 3.1 minutes. A sample of 72 customers who used a cashier averaged 6.1 minutes with a standard deviation of 2.8 minutes.

- Can you conclude that the mean checkout time is less for people who use the self-service lane?
- Can you conclude that if everyone used the self-service lane, that the mean checkout time would decrease?
Consider the number of items checked out when formulating your answer.

Solution:

Self service denoted by X.

$$n_x = 60 \quad \bar{x} = 5.2 \quad S_x = 3.1$$

$$n_y = 72 \quad \bar{y} = 6.1 \quad S_y = 2.8$$

a)

$$H_0: \mu_x - \mu_y = 0 \quad \text{versus} \quad H_1: \mu_x - \mu_y < 0$$

Under null hypothesis, we take $\mu_x - \mu_y = 0$. The null distribution of $\bar{X} - \bar{Y}$ is:

$$N(\mu_{\bar{x}} - \mu_{\bar{y}}, \sigma_{\bar{x}}^2 + \sigma_{\bar{y}}^2) = N(\mu_x - \mu_y, \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}) = N(0, 0.2691)$$

We observe $\bar{X} - \bar{Y} = 5.2 - 6.1 = -0.9$, therefore Z - score = $\frac{-0.9-0}{\sqrt{0.2691}} = -1.734945 \approx -1.73$

One-tailed test P-value is equal to 0.0418, then:

At 5% level we can reject null hypothesis, therefore $\mu_x < \mu_y$.

At 1% (or 2%) level we can not reject null hypothesis, therefore we can not conclude the mean μ_x is less than μ_y .

b)

While the results of the hypothesis tests suggest that self-service checkout lanes have significantly lower average checkout times compared to cashier checkout lanes, we cannot directly conclude that the mean checkout time would decrease if everyone used the self-service lane. This is because the number of items checked out by each customer can be an important factor that affects checkout time, and it may not be considered or controlled for in our analysis.

8)

In a study conducted by the U.S. Department of Health and Human Services, a sample of 546 boys aged 6–11 was weighed, and it was determined that 87 of them were overweight. A sample of 508 girls aged 6–11 was also weighed, and 74 of them were overweight. Can you conclude that the proportion of boys who are overweight differs from the proportion of girls who are overweight?

Solution:

$$H_0: P_b - P_g = 0 \quad \text{versus} \quad H_1: P_b - P_g \neq 0$$

$$n_b = 546 \quad X_b = 87 > 10 \quad n_b - X_b = 546 - 87 = 459 > 10$$

$$n_g = 508 \quad Y_g = 74 > 10 \quad n_g - Y_g = 508 - 74 = 434 > 10$$

All greater than 10, therefore traditional can be used!

$$\hat{P}_b = \frac{X_b}{n_b} = \frac{87}{546} = 0.159341 \quad \hat{P}_g = \frac{Y_g}{n_g} = \frac{74}{508} = 0.145669$$

$$\hat{P} = \frac{X_b + Y_g}{n_b + n_g} = \frac{87 + 74}{546 + 508} = \frac{161}{1054} = 0.152751$$

$$Z = \frac{\hat{P}_b - \hat{P}_g}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_b} + \frac{1}{n_g}\right)}} = \frac{0.159341 - 0.145669}{\sqrt{0.152751(1-0.152751)\left(\frac{1}{546} + \frac{1}{508}\right)}} = 0.616522 \approx 0.62 \rightarrow$$

$$P - \text{value} = 2(0.2676) = 0.5352 \approx 0.54 \rightarrow$$

P-value is not small enough to reject the null hypothesis. Then no, we cannot conclude that the proportion of boys who are overweight differs from the proportion of girls who are overweight.

9)

A new post-surgical treatment was compared with a standard treatment. Seven subjects received the new treatment, while seven others (the controls) received the standard treatment. The recovery times, in days, are given below.

Treatment: 12 13 15 19 20 21 24

Control: 18 23 24 30 32 35 39

Can you conclude that the mean recovery time for those receiving the new treatment is less than the mean for those receiving the standard treatment?

Solution:

X: New treatment

Y: Old treatment

$H_0: \mu_x - \mu_y \geq 0$

versus

$H_1: \mu_x - \mu_y < 0$

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}}$$

$$v = \frac{\left(\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}\right)^2}{\frac{\left(\frac{S_x^2}{n_x}\right)^2}{n_x - 1} + \frac{\left(\frac{S_y^2}{n_y}\right)^2}{n_y - 1}}$$

$$n_x = 7 \quad \bar{x} = 17.714286 \quad S_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n_x - 1}} = 4.461475$$

$$n_y = 7 \quad \bar{y} = 28.714286 \quad S_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n_y - 1}} = 7.387248$$

$$v = \frac{\left(\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}\right)^2}{\frac{\left(\frac{S_x^2}{n_x}\right)^2}{n_x - 1} + \frac{\left(\frac{S_y^2}{n_y}\right)^2}{n_y - 1}} = \frac{\left(\frac{4.461475^2}{7} + \frac{7.387248^2}{7}\right)^2}{\frac{(4.461475^2)^2}{6} + \frac{(7.387248^2)^2}{6}} = 9.863025 \rightarrow v = 9$$

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}} = \frac{(17.714286 - 28.714286) - (0)}{\sqrt{\frac{4.461475^2}{7} + \frac{7.387248^2}{7}}} = -3.37235 \approx -3.37$$

From the table, we find that the P-value is between 0.001 and 0.005. Therefore, we can reject the null hypothesis and conclude that $\mu_x < \mu_y$.

We also need to check whether the data sets are normal, we should at least check whether there is an outlier or not.

10)

The article “Determination of Carboxyhemoglobin Levels and Health Effects on Officers Working at the Istanbul Bosphorus Bridge” (G. Kocasoy and H. Yalin, *Journal of Environmental Science and Health*, 2004:1129–1139) presents assessments of health outcomes of people working in an environment with high levels of carbon monoxide (CO). Following are the numbers of workers reporting various symptoms, categorized by work shift. The numbers were read from a graph.

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	Shift		
	Morning	Evening	Night
Influenza	16	13	18
Headache	24	33	6
Weakness	11	16	5
Shortness of Breath	7	9	9

Can you conclude that the proportions of workers with the various symptoms differ among the shifts?

Solution:

Observed Values	Morning	Evening	Night	Total
Influenza	16	13	18	47
Headache	24	33	6	63
Weakness	11	16	5	32
Shortness of Breath	7	9	9	25
<u>Total</u>	58	71	38	167

H_0 : Proportions of workers with the various symptoms are independent from the shifts.

H_1 : Proportions of workers with the various symptoms are not independent from the shifts.

For example, the expected number of influenza cases in the morning can be calculated by multiplying as follows:

Expected number of Influenza-Morning :

$$\frac{\text{Number of Influenza}}{\text{Total Number}} \times \frac{\text{Number in Morning}}{\text{Total Number}} \times \text{Total Number} = \frac{47}{167} \times \frac{58}{167} \times 167 = 16.323$$

Expected Values	Morning	Evening	Night	Total
Influenza	16.323	19.982	10.695	47
Headache	21.880	26.784	14.335	63
Weakness	11.114	13.605	7.281	32
Shortness of Breath	8.683	10.629	5.689	25
<u>Total</u>	58	71	38	167

$$\chi^2 = \sum_{i=1}^4 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$\chi^2 = \frac{(16 - 16.323)^2}{16.323} + \frac{(13 - 19.982)^2}{19.982} + \frac{(18 - 10.695)^2}{10.695} + \dots + \frac{(9 - 10.629)^2}{10.629} + \frac{(9 - 5.689)^2}{5.689} = 17.571553$$

The number of degrees of freedom is $(4-1)*(3-1)=6$, then P-value is between 0.005 and 0.01 → Yes, it is reasonable to conclude that the proportions of workers with various symptoms differ among the shifts.

11)

Scores on the math SAT are normally distributed. A sample of 20 SAT scores had standard deviation $s = 87$. Someone says that the scoring system for the SAT is designed so that the population standard deviation will be $\sigma = 100$. Do these data provide sufficient evidence to contradict this claim?

Solution:

$$n = 20 \quad S = 87$$

$$H_0: \sigma = 100 \quad \text{versus} \quad H_1: \sigma \neq 100$$

$$\text{value of test statistic: } \frac{(n-1)s^2}{\sigma^2} = \frac{19(87)^2}{100^2} = 14.3811$$

Then, $P\text{-value} > 0.1 \rightarrow$ we can not reject that claim (null hypothesis).

12)

A broth used to manufacture a pharmaceutical product has its sugar content, in mg/mL, measured several times on each of three successive days.

Day 1: 5.0 4.8 5.1 5.1 4.8 5.1 4.8

4.8 5.0 5.2 4.9 4.9 5.0

Day 2: 5.8 4.7 4.7 4.9 5.1 4.9 5.4

5.3 5.3 4.8 5.7 5.1 5.7

Day 3: 6.3 4.7 5.1 5.9 5.1 5.9 4.7

6.0 5.3 4.9 5.7 5.3 5.6

- Can you conclude that the variability of the process is greater on the second day than on the first day?
- Can you conclude that the variability of the process is greater on the third day than on the second day?

Solution:

$$F_{n_i-1, n_j-1} = \frac{S_i^2}{S_j^2}$$

$$n_1 = n_2 = n_3 = 13 \quad S_1 = 0.138675 \quad S_2 = 0.384808 \quad S_3 = 0.518256$$

a)

$$H_0: \sigma_2 \leq \sigma_1 \quad \text{versus} \quad H_1: \sigma_2 > \sigma_1$$

$$F_{12,12} = \frac{S_2^2}{S_1^2} = 7.7 \quad \text{The null distribution of the test is } F_{12,12} = 7.7, \text{ from Table A.8:}$$

P-value $< 0.001 \rightarrow$ null hypothesis is rejected and we can conclude that $\sigma_2 > \sigma_1$.

b)

$$H_0: \sigma_3 \leq \sigma_2 \quad \text{versus} \quad H_1: \sigma_3 > \sigma_2$$

$$F_{12,12} = \frac{S_3^2}{S_2^2} = 1.813853 \approx 1.81 \rightarrow \text{P-value} > 0.1 \rightarrow \text{we can not reject null hypothesis.}$$

13)

Scores on a certain IQ test are known to have a mean of 100. A random sample of 60 students attend a series of coaching classes before taking the test. Let μ be the population mean IQ score that would occur if every student took the coaching classes. The classes are successful if $\mu > 100$. A test is made of the hypotheses $H_0: \mu \leq 100$ versus $H_1: \mu > 100$.

- Assume that the classes are successful but the conclusion is reached that the classes might not be successful. Which type of error is this?
- Assume that the classes are not successful. Is it possible to make a Type I error? Explain.
- Assume that the classes are not successful. Is it possible to make a Type II error? Explain.

Solution:

$$H_0: \mu \leq 100 \quad \text{versus} \quad H_1: \mu > 100 \quad n=60$$

a) Based on the test result, we cannot reject the null hypothesis, which suggests that the classes might not be successful. However, in reality, the classes are successful. This situation represents a Type II error, which occurs when we fail to reject a false null hypothesis.

b) If the conclusion is reached that the classes are successful when they are not, then we will have a Type I error. A Type I error occurs when we reject a true null hypothesis.

c) It is impossible, since the assumption is that the null hypothesis is not false.